CS 70 Discrete Mathematics and Probability Theory
Fall 2025 Sabin, Hug HW 00

1 Administrivia

(a) Make sure you are on the course Ed (for Q&A) and Gradescope (for submitting homeworks, including this one). Find and familiarize yourself with the course website. What is its homepage's URL?

(b) Read the policies page on the course website.

(i) What is the breakdown of how your grade is calculated?

(ii) What is the attendance policy for discussions?

(iii) When are homeworks released, and when are they due?

(iv) How many "drops" do you get for homeworks? How many mini-vitamins will contribute to your grade?

(v) When is the midterm? When is the final?

(vi) What percentage score is needed to earn full credit on a homework?

Solution:

(a) The course website is located at `https://www.eecs70.org/`.

(b) (i) For HW option: Discussion Attendance: 5%, Mini-vitamins: 5%, Homework: 15%, Midterm: 30%, Final: 45%.

For no HW option: Discussion Attendance: 5%, Mini-vitamins: 5%, Midterm: 36%, Final: 54%.

(ii) You will receive 1 attendance point for every discussion, and will need at least 13 points in order to receive full credit for discussion attendance. You are welcome to attend other discussion sections, and may receive credit for any section, though it is *strongly encouraged* that you consistently attend the discussion that you sign up for at the beginning of the semester.

(iii) The homework for the current week is released on the course website on Saturday. The homework is due on Gradescope the following Saturday at 4:00 PM (grace period until 6:00 PM); the solutions for that homework will be released on Monday, a day after the release of the new homework.

(iv) You can drop the lowest 3 homeworks the entire semester, and the top 13 mini vitamins will count for a grade. However, please save these drops for emergencies. We do not have the bandwidth to make personalized exceptions to this rule.

CS 70 离散数学与概率论

2025 年秋季 萨宾，拥抱 HW 00

1 行政琐事

（a）请确保您已登录课程 Ed（用于问答）和 Gradescope（用于提交作业，包括本次作业）。找到并熟悉课程网站。它的主页网址是什么？

（b）阅读课程网站上的政策页面。

（i）你的成绩是如何计算的？

（二）讨论课的出席政策是什么？

（三）作业何时发布，何时截止？

（四）作业能给你多少“分数”？多少“小维生素”能帮你提高成绩？

（五）期中考试是什么时候？期末考试是什么时候？

（六）作业需要达到多少百分比的分数才能获得满分？

解决方案：

（a）课程网站位于 https://www.eecs70.org/。

（b）（i）对于作业选项：讨论出勤率：5%，迷你维生素：5%，作业：15%，期中考试：30%，期末考试：45%。

对于不布置作业的情况：讨论课出勤率：5%，迷你维生素：5%，期中考试：36%，期末考试：54%。

(ii) 每次讨论课您将获得 1 个出勤积分，至少需要 13 个积分才能获得讨论课出勤的满分。欢迎您参加其他讨论课，并且任何讨论课的出勤积分均可获得，但我们强烈建议您始终参加学期初报名的讨论课。

（三）本周作业将于周六在课程网站上发布。作业需在下周六下午 4:00 前通过 Gradescope 提交（宽限期至下午 6:00）；该作业的答案将于下周一发布，即新作业发布后的第二天。

(iv) 你可以放弃整个学期最低的 3 次作业成绩，成绩最高的 13 份迷你维生素（可能是指某种营养补充剂或营养补充剂）将计入最终成绩。但是，请仅在紧急情况下才使用这些放弃作业的机会。我们没有足够的资源来为个人特殊情况破例。

(v) Midterm Date: 10/16/25 Thursday between 7–10pm

Final Date: 12/19/25 Friday 7–10pm

(vi) 73%.

2 Course Policies

Go to the course website and read the course policies carefully. Leave a followup on Ed if you have any questions. Are the following situations violations of course policy? Write "Yes" or "No", and a short explanation for each.

(a) Alice and Bob work on a problem in a study group. They write up a solution together and submit it, noting on their submissions that they wrote up their homework answers together.

(b) Carol goes to a homework party and listens to Dan describe his approach to a problem on the board, taking notes in the process. She writes up her homework submission from her notes, crediting Dan.

(c) Erin comes across a proof that is part of a homework problem while studying course material. She reads it and then, after she has understood it, writes her own solution using the same approach. She submits the homework with a citation to the website.

(d) Frank is having trouble with his homework and asks Grace for help. Grace lets Frank look at her written solution. Frank copies it onto his notebook and uses the copy as a reference to write and submit his homework, crediting Grace.

(e) Heidi has completed her homework using LaTeX. Her friend Irene has been working on a homework problem for hours, and asks Heidi for help. Heidi sends Irene her PDF solution, and Irene uses it to write her own solution with a citation to Heidi.

(f) Joe found homework solutions before they were officially released, and every time he got stuck, he looked at the solutions for a hint. He then cited the solutions as part of his submission.

(g) Kai is struggling with one of their homework problems, so they take a screenshot of the problem and ask ChatGPT to solve it for them. They adapt ChatGPT's response to their own solution, and they include a link to their ChatGPT conversation in the homework's Sundry section.

Solution:

(a) Yes, this is a violation of course policy. All solutions must be written entirely by the student submitting the homework. Even if students collaborate, each student must write a unique, individual solution. In this case, both Alice and Bob would be culpable.

(b) No, this is not a violation of course policy. While sharing *written solutions* is not allowed, sharing *approaches* to problems is allowed and encouraged. Because Carol only copied down *notes*, not *Dan's solution*, and properly cited Dan's contribution, this is an actively encouraged form of collaboration.

（五）期中考试日期：2025 年 10 月 16 日星期四晚上 7 点至 10 点

截止日期：2025 年 12 月 19 日星期五晚上 7 点至 10 点

（六）73%。

2 课程政策

请访问课程网站并仔细阅读课程政策。如有任何疑问，请在 Ed 上留言。以下情况是否违反课程政策？请填写“是”或“否”，并简要解释每种情况。

（a）爱丽丝和鲍勃在一个学习小组中一起解决一个问题。他们共同写出解决方案并提交，并在提交的作业中注明他们是一起完成作业答案的。

(b) 卡罗尔去参加一个家庭作业辅导会，听丹在黑板上讲解他解决问题的方法，并记下了笔记。她根据笔记完成了自己的作业，并在作业中注明了丹的解题思路。

（c）艾琳在学习课程材料时，偶然发现了一道家庭作业题的证明。
她阅读了原文，理解之后，用同样的方法写出了自己的解答。她提交作业时，附上了网站的出处。

(d) 弗兰克做作业遇到困难，便向格蕾丝求助。格蕾丝让弗兰克看了她写好的答案。弗兰克把答案抄到笔记本上，并以此为参考完成作业，提交时注明了格蕾丝的贡献。

(e) 海蒂用 LTEX 完成了作业。她的朋友艾琳做了好几个小时的作业题，于是向海蒂求助。海蒂把她的 PDF 解答发给了艾琳，艾琳根据海蒂的解答写出了自己的答案，并注明了出处。

(f) 乔在作业答案正式发布前就找到了答案，每次遇到难题，他都会参考答案寻找线索。然后，他在提交的作业中引用了这些答案。

(g) Kai 在做作业时遇到一道难题，于是他截屏了题目，并请 ChatGPT 帮忙解答。他根据 ChatGPT 的解答修改了自己的答案，并将 ChatGPT 的对话链接添加到了作业的“其他”部分。

解决方案：

(a) 是的，这违反了课程规定。所有作业答案必须完全由提交作业的学生独立完成。即使学生合作，每个学生也必须撰写一份独一无二的独立答案。在这种情况下，Alice 和 Bob 都应承担责任。

(b) 不，这并未违反课程规定。虽然不允许分享书面答案，但允许并鼓励分享解题思路。由于卡罗尔只是抄写了笔记，而非丹的答案，并且正确引用了丹的贡献，因此这是一种积极鼓励的合作形式。

(c) No, this is not a violation of course policy. Using external sources to help with homework problems, while less encouraged than peer collaboration, is fine as long as (i) the student makes sure to understand the solution; (ii) the student uses understanding to write a new solution, and does not copy from the external source; and (iii) the student credits the external source. However, looking up a homework problem online is a violation of course policies; the correct course of action upon finding homework solutions online is to close the tab.

(d) Yes, this is a violation of course policy, and both Frank and Grace would be culpable. Even though Frank credits Grace, written solutions should never be shared in the first place, and certainly not copied down. This is to ensure that each student learns how to write and present clear and convincing arguments. To be safe, try not to let anybody see your written solutions at any point in the course—restrict your collaboration to *approaches* and *verbal communication*.

(e) Yes, this is a violation of course policy. Once again, a citation does not make up for the fact that written solutions should never be shared, in written or typed form. In this case, both Heidi and Irene would be culpable.

(f) Yes, this is a violation of course policy. Joe should not be reading solutions before they are officially released. Instead, Joe should ask for help when he is stuck through Ed or Office Hours.

(g) Yes, this is a violation of course policy. Kai should not be directly asking ChatGPT to solve a homework question, despite correctly including the conversation history in their Sundry. If Kai really wanted to use an LLM for homework help, they should ask it a conceptual question (that they'd reasonably ask at Office Hours or on Ed).

3 Use of Ed

Ed is incredibly useful for Q&A in such a large-scale class. We will use Ed for all important announcements. You should check it frequently. We also highly encourage you to use Ed to ask questions and answer questions from your fellow students.

(a) Read the Ed Etiquette section of the course policies and explain what is wrong with the following hypothetical student question: "Can someone explain the proof of Theorem XYZ to me?" (Assume Theorem XYZ is a complicated concept.)

(b) When are the weekly posts released? Are they required reading?

(c) If you have a question or concern not directly related to the course content, where should you direct it?

Solution:

(a) There are two things wrong with this question. First, this question does not pass the 5-minute test. This concept takes longer than 5 minutes to explain, and therefore is better suited to Office Hours. Second, this question does not hone in on a particular concept with which the student is struggling. Questions on Ed should be narrow, and should include every step of

(c) 不，这并不违反课程规定。虽然使用外部资源来帮助完成作业题不如同伴合作那么鼓励，但只要满足以下条件，使用外部资源是可以的：(i) 学生确保理解了答案；(ii) 学生运用理解写出了新的答案，而不是抄袭外部资源；(iii) 学生注明了外部资源的出处。然而，在网上查找作业题答案是违反课程规定的；在网上找到作业答案后，正确的做法是立即关闭网页。

(d) 是的，这违反了课程规定，Frank 和 Grace 都应承担责任。即使 Frank 承认 Grace 的贡献，书面解答也绝对不应该分享，更不应该抄袭。这样做是为了确保每位学生都能学会如何撰写和陈述清晰且令人信服的论点。为了安全起见，在课程的任何阶段都不要让任何人看到你的书面解答——你们的合作仅限于思路探讨和口头交流。

(e) 是的，这违反了课程规定。再次强调，开罚单并不能弥补书面解答绝对不能以任何形式（包括手写或打印形式）分享的事实。在这种情况下，海蒂和艾琳都应承担责任。

(f) 是的，这违反了课程规定。乔不应该在答案正式发布前就阅读答案。相反，乔应该在遇到问题时通过 Ed 或答疑时间寻求帮助。

(g) 是的，这违反了课程规定。尽管 Kai 正确地将对话记录添加到了他的杂项列表中，但他不应该直接要求 ChatGPT 解答作业题。如果 Kai 真的想用 LLM 来获得作业帮助，他应该问一些概念性的问题（比如他会在答疑时间或 Ed 上合理提出的问题）。

3 埃德的使用

在这样的大型课堂上，Ed 对于问答环节非常有用。我们将使用 Ed 发布所有重要通知，请大家经常查看。我们也强烈建议大家使用 Ed 向同学提问和回答问题。

（a）阅读课程政策中的“教育礼仪”部分，并解释以下假设的学生问题有何不妥之处：“有人能给我解释一下定理 XYZ 的证明吗？”（假设定理 XYZ 是一个复杂的概念。）

（b）每周的文章何时发布？是必读的吗？

（c）如果您有与课程内容没有直接关系的问题或疑虑，您应该向谁咨询？

解决方案：

(a) 这个问题有两个问题。首先，这个问题不符合五分钟测试的要求。解释这个概念需要的时间超过五分钟，因此更适合在答疑时间提出。其次，这个问题没有针对学生正在努力理解的特定概念。Ed 上的问题应该切题，并且应该涵盖每个步骤。

reasoning that led up to the question. A better question in this case might be: "I understood every step of the proof of Theorem XYZ in Note 2, except for the very last step. I tried to reason it like this, but I didn't see how it yielded the result. Can someone explain where I went wrong?"

(b) The weekly posts are released every Monday. They're required reading.

(c) Please send an email to `fa25@eecs70.org`.

4 Academic Integrity

Please write or type out the following pledge in print, and sign it.

> I pledge to uphold the university's honor code: to act with honesty, integrity, and respect for others, including their work. By signing, I ensure that all written homework I submit will be in my own words, that I will acknowledge any collaboration or help received, and that I will neither give nor receive help on any examinations.

5 Propositional Practice

Note 1

In parts (a)–(b), convert the English sentences into propositional logic. In parts (c)–(d), convert the propositions into English. For parts (b) and (d), use the notation $a \mid b$ to denote the statement "a divides b", and use the notation $P(x)$ to denote the statement "x is a prime number".

(a) For every real number k, there is a unique real solution to $x^3 = k$.

(b) If p is a prime number, then for any two natural numbers a and b, if p doesn't divide a and p divides ab, then p divides b.

(c) $(\forall x, y \in \mathbb{R})[(xy = 0) \implies ((x = 0) \vee (y = 0))]$

(d) $\neg((\exists y \in \mathbb{N})[(\forall x \in \mathbb{N})[(x > y) \implies ((y \mid x) \vee P(x))]])$

Solution:

(a) The trickiest part of this problem is the word 'unique'. We can express the existence of a unique solution in propositional logic with two statements connected with an 'and': (1) A solution exists, and (2) Any two solutions have to be the same. Hence, we can rewrite this statement as "For every real number k, there exists a real number x such that $x^3 = k$ and for all reals y and z, if both $y^3 = k$ and $z^3 = k$, then $y = z$." This, in propositional logic, is below:

$$(\forall k \in \mathbb{R})\big[(\exists x \in \mathbb{R})(x^3 = k) \wedge (\forall y, z \in \mathbb{R})(((y^3 = k) \wedge (z^3 = k)) \implies (y = z))\big].$$

(b) This sentence can be written in propositional logic as

$$(\forall p \in \mathbb{N})[(P(p)) \implies ((\forall a, b \in \mathbb{N})[((p \mid ab) \wedge \neg(p \mid a)) \implies (p \mid b)])].$$

(c) If the product of two real numbers is 0, then one of them must be 0.

(d) There is no natural number that divides every composite number greater than it.

提出这个问题之前的推理过程。更好的问题可能是： “我理解了注释 2 中定理 XYZ 证明的每一步，除了最后一步。我尝试按照这个思路推理，但我不明白它是如何得出结果的。请问有人能解释一下我哪里出错了？”

（b）每周一都会发布周刊文章，是必读文章。

（c）请发送电子邮件至 fa25@eecs70.org。

4 学术诚信

请将以下承诺以书面或打字形式写下来并签名。

> 我承诺遵守大学的荣誉准则：诚实守信，尊重他人，包括他人的作品。我在此签字，保证我提交的所有书面作业均由我本人撰写，我会注明所有合作或获得的帮助，并且不会在任何考试中给予或接受帮助。

5 命题实践

注 1

在 (a) 至 (b) 部分，将英文句子转换为命题逻辑。在 (c) 至 (d) 部分，将命题转换为英文。对于 (b) 和 (d) 部分，使用符号 $a \mid b$ 表示命题“a 整除 b”，使用符号 $P(x)$ 表示命题“x 是素数”。

（a）对于每个实数 k，方程 $x=k$ 都有唯一的实数解。

（b）如果 p 是质数，那么对于任意两个自然数 a 和 b，如果 p 不能整除 a 且 p 能整除 ab，那么 p 能整除 b。

(c) $(\forall x, y \in \mathbb{R})[(xy = 0) \implies ((x = 0) \vee (y = 0))]$

(d) $\neg((\exists y \in \mathbb{N})[(\forall x \in \mathbb{N})[(x > y) \implies ((y \mid x) \vee P(x))]])$

解决方案：

(a) 这个问题最棘手的部分在于“唯一”这个词。我们可以用命题逻辑中的两个由“与”连接的语句来表达唯一解的存在性：(1) 存在一个解，并且 (2) 任意两个解必须相同。因此，我们可以将这个语句重写为“对于每个实数 k，存在一个实数 x，使得 $x = k$ 且对于所有

> 对于所有实数 y 和 z，如果 $y = k$ 且 $z = k$，则 $y = z$。” 用命题逻辑表示如下：

$$(\forall k \in \mathbb{R})(\exists x \in \mathbb{R})(x = k) \wedge (\forall y, z \in \mathbb{R})(((y = k) \wedge (z = k)) \implies (y = z)) \quad .$$

(b) 这句话可以用命题逻辑写成：

$$(\forall p \in \mathbb{N})[(P(p)) \implies ((\forall a, b \in \mathbb{N})[((p \mid ab) \wedge \neg(p \mid a)) \implies (p \mid b)])].$$

（c）如果两个实数的乘积为 0，那么其中一个实数必定为 0。

（d）不存在一个自然数能整除比它大的所有合数。